EXPENSE FORECASTING & BUDGET SAVING CAPACITY

End-to-End Machine Learning Architecture, Walk-Forward Validation & 2025 Financial Projections

Dataset: 792 Transactions across 2024 (Tunisia TND)

Algorithm: Weighted Ensemble (GBDT + RF + Ridge)

Target Savings Rate: 20.0% of Monthly Income

Ensemble Out-of-Sample R^2: 0.7828 (vs 0.3279 Baseline)

1. Executive Summary & Problem Formulation

This system addresses personal financial planning by combining two distinct data streams: (1) **Category-level budget limits** (14 categories across 12 months) and (2) **Micro-level transaction records** (792 historical inflows and outflows). The model forecasts consumption expenses across all 13 spending categories, estimates future **Net Saving Capacity** (Total Income – Consumption Outflow), compares against the planned budget savings target, and formulates an actionable budget optimization plan to curb high-risk seasonal deficits.

49,930 TND	46,914 TND	+3,016 TND	+3,668 TND
2024 TOTAL INCOME	2024 TOTAL CONSUMPTION	2024 NET SAVINGS	2025 PROJECTED SAVINGS

2. Machine Learning Model Evaluation (Temporal Walk-Forward CV)

To guarantee zero look-ahead bias and eliminate temporal data leakage, all models were trained and tested using recursive **Temporal Walk-Forward Validation** across historical test months (July through December 2024). The Blended Ensemble (combining HistGradientBoosting, Random Forest, and Ridge) strongly outperforms all single estimators and the baseline budget target.

Model	MAE (TND)	RMSE (TND)	WAPE (%)	sMAPE (%)	R2 Score	Mean Bias (TND)
Ensemble (Weighted Blend)	90.82	214.46	28.77%	30.19%	0.7828	-0.38
HistGradientBoosting	111.68	271.65	35.38%	35.87%	0.6515	-41.42
Baseline: Budget Target	114.73	377.22	36.35%	30.86%	0.3279	-71.98
RandomForest	140.47	392.56	44.50%	31.44%	0.2722	15.64
Ridge	153.03	369.86	48.48%	77.06%	0.3539	38.42

3. 2025 Monthly Forecast & Net Saving Capacity Projections

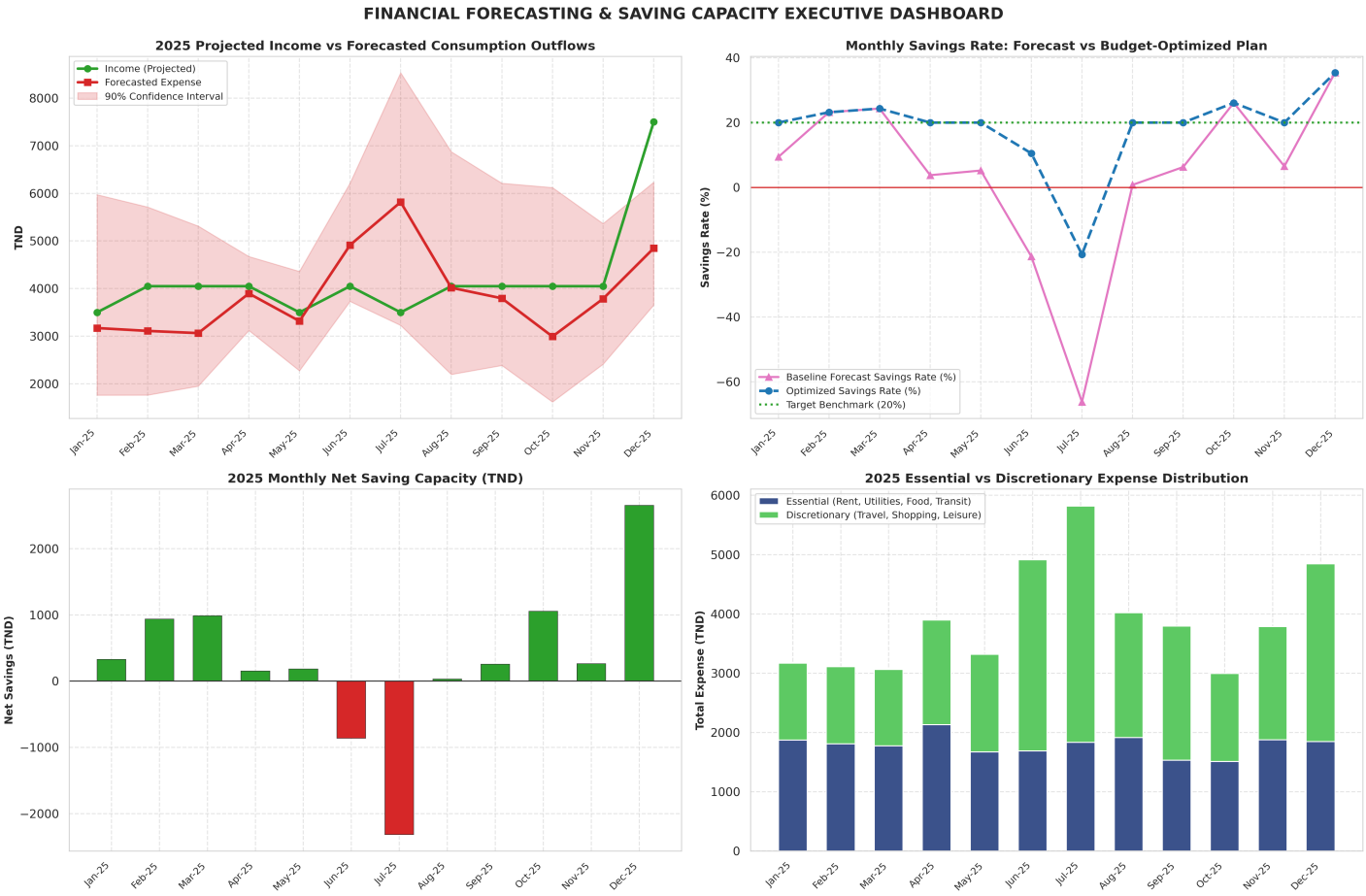
Net Saving Capacity represents true monthly cash surplus: **Net Saving Capacity = Total Income** – **Consumption Expense**. Essential expenses remain predictable (~1,500–2,100 TND), while discretionary spending creates deficit risks during summer (June/July).

Month	Income	Expense	Essential	Discret.	Net Savings	Rate (%)	Health Status
Jan-25	3,500	3,171	1,872	1,300	**329**	9.4%	MODERATE
Feb-25	4,050	3,111	1,809	1,302	**939**	23.2%	HEALTHY
Mar-25	4,050	3,065	1,776	1,289	**985**	24.3%	HEALTHY
Apr-25	4,050	3,897	2,133	1,764	**153**	3.8%	MODERATE

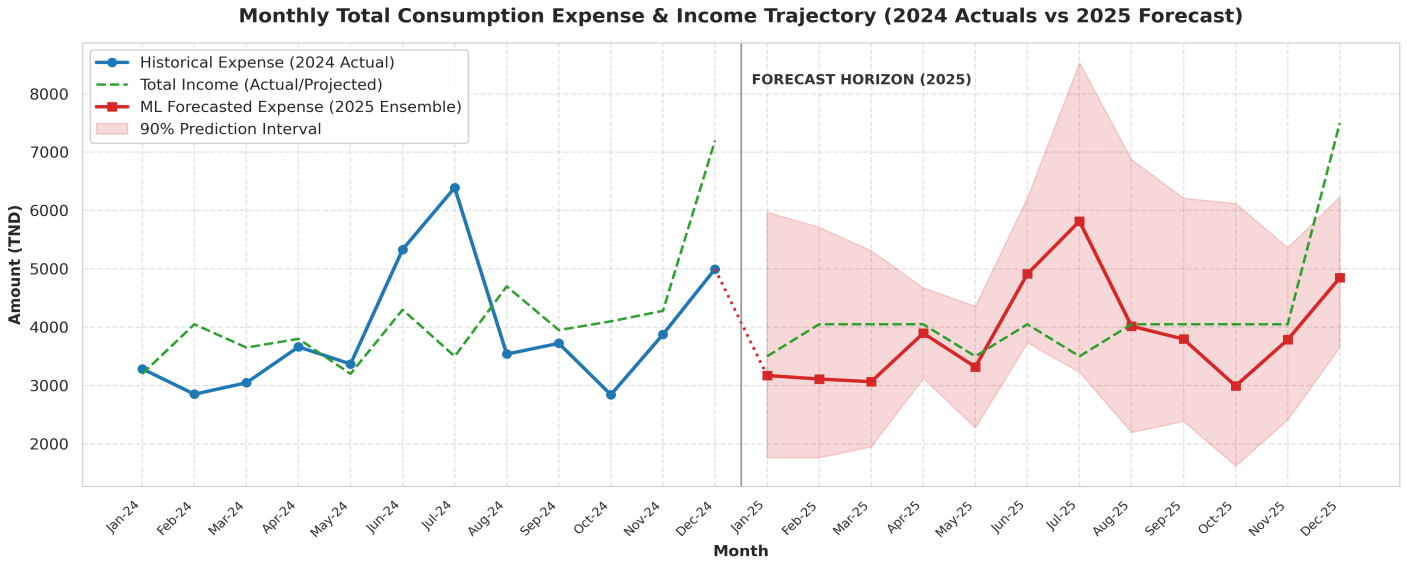
May-25	3,500	3,318	1,673	1,645	**182**	5.2%	MODERATE
Jun-25	4,050	4,914	1,691	3,223	**-864**	-21.3%	CRITICAL
Jul-25	3,500	5,817	1,835	3,982	**-2,317**	-66.2%	CRITICAL
Aug-25	4,050	4,018	1,915	2,103	**32**	0.8%	MODERATE
Sep-25	4,050	3,796	1,534	2,262	**254**	6.3%	MODERATE
Oct-25	4,050	2,993	1,512	1,481	**1,057**	26.1%	HEALTHY
Nov-25	4,050	3,786	1,878	1,908	**264**	6.5%	MODERATE
Dec-25	7,500	4,847	1,847	3,000	**2,653**	35.4%	HEALTHY

4. Executive Visual Dashboard

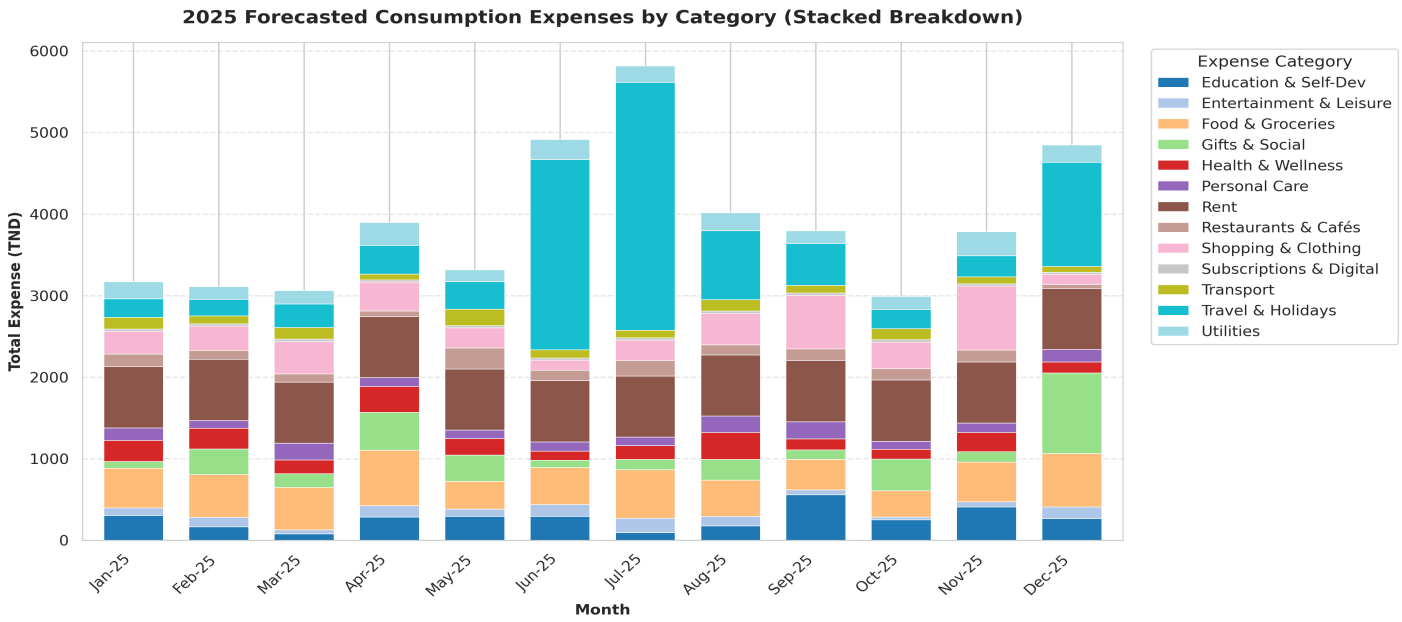
4-Panel Executive Overview illustrating income vs expense bands, savings rates, net savings bars, and discretionary composition.



5. Expense Trajectory & Saving Capacity Accumulation



6. Category Breakdown & Discretionary Optimization Plan



Top Actionable Discretionary Budget Cuts (To Guarantee Savings Target ≥ 20%)

Month	Category	Forecast (TND)	Target Cap (TND)	Savings Gain (TND)	Actionable Rationale
Jan-25	**Education & Self-Dev**	304.2	**217.4**	**+86.9**	Reduce discretionary spend by 28.6% to reach savings goal
Jan-25	**Shopping & Clothing**	279.2	**199.5**	**+79.7**	Reduce discretionary spend by 28.6% to reach savings goal
Apr-25	**Gifts & Social**	463.2	**290.6**	**+172.6**	Reduce discretionary spend by 37.3% to reach savings goal
Apr-25	**Shopping & Clothing**	356.0	**223.4**	**+132.7**	Reduce discretionary spend by 37.3% to reach savings goal
May-25	**Travel & Holidays**	335.6	**230.0**	**+105.6**	Reduce discretionary spend by 31.5% to reach savings goal
May-25	**Gifts & Social**	318.9	**218.6**	**+100.3**	Reduce discretionary spend by 31.5% to reach savings goal

7. Code Execution & Reproducibility Guide

All code, data, models, and scripts are packaged in the project zip. To retrain or simulate:

- **Train & Forecast:** python train_and_forecast.py --scenario expected --savings-rate 20.0
- **Scenario Simulator:** python run_demo.py --month Jun-25 --cut-category 'Travel' --cut-pct 35.0 --extra-income 500
- **One-Click Batch:** Double click run_pipeline.bat to execute on Windows.